AWS EC2 Web Server Deployment using Nginx

Name

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Objective

The objective of this lab is to launch an Ubuntu EC2 instance on AWS, connect using SSH, install and configure the Nginx web server, host a simple HTML website, upload the project to GitHub, and understand basic Linux administration commands.

Task 1: AWS EC2 Setup

AWS Services Used

- Amazon EC2
- Security Group
- Key Pair
- VPC
- Internet Gateway

Steps Performed

- Launched an Ubuntu EC2 instance.
- Created a Security Group.
- Allowed inbound ports:
    - SSH (22)
    - HTTP (80)
- Connected to the EC2 instance using SSH.

Task 2: Linux Basics

Commands Used

sudo apt update

sudo apt upgrade -y

sudo apt install nginx -y

sudo systemctl status nginx

sudo systemctl restart nginx

df -h

free -h

ps -ef

Purpose

- Updated system packages.
- Installed the Nginx web server.
- Verified Nginx service status.
- Restarted the service.
- Checked disk usage.
- Checked memory usage.
- Viewed running processes.

Task 3: Website Deployment

Steps Performed

- Removed the default Nginx page.
- Created a new index.html file.
- Added:
    - Name
    - College
    - Branch
    - Email
    - Current Date
- Restarted Nginx.
- Verified the website using the EC2 Public IP.

Task 4: Git & GitHub

Steps Performed

- Created a GitHub repository.
- Added the HTML file and README.md.
- Committed the project.
- Pushed the files to GitHub.

Git Commands

git init

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git add .

git commit -m "Added files to repo"

git push origin main
```

Problems Faced

- HTTP port was initially not accessible because port 80 was not allowed in the Security Group.
- Faced permission issues while editing files and resolved them using sudo.

Learnings

- Launching and managing an EC2 instance.
- Connecting to Linux using SSH.
- Installing and managing Nginx.
- Hosting a static website.
- Using basic Linux commands.
- Uploading a project to GitHub.

Time Taken

Approximately **2 hours** to complete the lab.

Bonus Task – Attach Elastic IP

Objective:
Attach an Elastic IP address to the EC2 instance so that it has a static public IP.

Steps Performed:

1. Opened the EC2 Dashboard.
2. Navigated to **Elastic IPs**.
3. Allocated a new Elastic IP address.
4. Associated the Elastic IP with the running EC2 instance.